

Executive Summary

Radiomics-genomics association studies require specialized metrics beyond simple linear correlation to capture complex relationships and evaluate predictive models. Key approaches include univariate tests (t-test, ANOVA, χ^2) and effect sizes (Cohen's d, odds ratio), classification metrics (ROC AUC, accuracy, F1, MCC), information-theoretic measures (mutual information, entropy), and multivariate or model-based analyses (logistic/Cox regression, LASSO, random forest, SHAP). Each metric has specific assumptions (e.g. linearity, distributional form), strengths (e.g. interpretability, nonlinearity, handling high-dimensionality) and limitations (e.g. sensitivity to class imbalance or confounding). For radiogenomics, confounders like scanner differences, segmentation variability and tumor volume can bias associations; preprocessing (intensity normalization, ComBat harmonization, feature z-scoring) and covariate adjustment are critical. Robust validation (cross-validation, external cohorts, bootstrapping, permutation tests, multiple-testing correction) is essential to avoid overfitting and false positives.

We recommend a pipeline with rigorous feature preprocessing/harmonization, exploratory univariate screening, multivariate modeling (with regularization and feature importance analysis), and thorough validation [53†L5844-L5847] [38†L193-L202]. For discovery studies, prioritize interpretable effect-size measures (Cohen's d, OR) and mutual information to flag candidate features; for predictive modeling, emphasize classification metrics (AUC) and model-agnostic importance (SHAP) with nested CV; for causal hypotheses, adjust for confounders via multivariable models or DAG-based methods and consider experimental validation [42†L2775-L2784] [53†L5844-L5847]. The table below compares common metrics. Example code (Python/pseudocode) illustrates computation of mutual information, bootstrap AUC CIs, logistic ORs, LASSO coefficients, SHAP values, and the survival C-index.

flowchart TB

```
A[Collect imaging & genomics data] --> B[Segmentation & preprocessing]
B --> C[Feature extraction (radiomics)]
C --> D[Harmonize & normalize features]
D --> E[Univariate analysis (t-test, MI, OR, etc.)]
D --> F[Multivariate modeling (LR, RF, Cox, etc.)]
E & F --> G{Validation}
G --> G1[Cross-validation & bootstrap]
G --> G2[External cohort testing]
G --> G3[Permutation & multiple-testing correction]
G1 & G2 & G3 --> H[Model interpretation (effect sizes, SHAP)]
H --> I[Integration with clinical outcomes]
I --> J[Reporting & clinical translation]
```

Statistical Association Tests (Univariate)

Pearson correlation (r) measures linear association between a continuous feature and a continuous/genotypic outcome (e.g. mutation burden). It assumes normally distributed variables and linearity; it is *not*

robust to outliers or nonlinear relations. The formula is $r = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sqrt{\sum(x-\bar{x})^2 \sum(y-\bar{y})^2}}$. In radiogenomics, Pearson r is simple but sensitive to confounders and feature scaling. Its effect size is directly r ; interpretation is percent variance explained (R^2). **Spearman rank correlation** is a nonparametric analog (calculates Pearson r on ranks) that handles monotonic but nonlinear associations (less powerful for linear patterns but more robust to outliers and non-normality).

T-test / ANOVA compare feature distributions between mutation+ vs mutation- groups. For a binary mutation, an independent two-sample t-test (or nonparametric Mann-Whitney U) tests difference of means; ANOVA generalizes to >2 groups (e.g. different mutation subtypes). Assumptions include independent samples and equal variance (Welch's t-test relaxes equal-variance). The test statistic yields a p -value for association. Strength: straightforward and well-understood; limitation: ignores covariates/confounders, multiple testing issues if many features. **Effect size:** Cohen's d (standardized mean difference) {

$$34\uparrow L41 - L49$$

};

$$d = \frac{\bar{x}_1 - \bar{x}_2}{SD_{\text{pooled}}}, \quad SD_{\text{pooled}} = \sqrt{\frac{(n_1 - 1) SD_1^2 + (n_2 - 1) SD_2^2}{n_1 + n_2 - 2}}.$$

Cohen's benchmarks: 0.2=small, 0.5=medium, 0.8=large **[34\uparrow L79-L87]** . In radiogenomics, d is useful to gauge practical significance; small p -values with tiny d (often in big datasets) may be unimportant.

Chi-square / Fisher's exact test apply when features or outcomes are categorical. E.g., discretizing a radiomic feature or classifying tumors by texture type vs mutation presence. Yates-corrected χ^2 or Fisher's exact test give p -values; effect sizes include Cramer's V or odds ratio (for 2x2 tables). These are seldom used for truly continuous radiomics data, but can arise after binning or clustering. They share limitations with t -tests regarding confounding and sample size (cells with low counts reduce power).

Preprocessing: For t -tests/correlation, radiomic features are often z -scored or normalized (e.g. feature-mean/SD) to ensure comparability and avoid dominance by scale. Tumor size (volume) is a confounder: if larger tumors tend to have certain mutations, size-adjusted comparisons (ANCOVA or including size as a covariate) may be needed. Segmentation variability can inflate within-group variance, reducing test power **[38\uparrow L193-L202]** . Nonparametric tests (Wilcoxon, Kruskal-Wallis) avoid normality assumption but still assume independent samples.

Multiple testing: With thousands of radiomic features, many univariate tests are performed. Apply corrections (Benjamini-Hochberg FDR or Bonferroni) to control false positives. Permutation testing (shuffling mutation labels) can estimate empirical p -values.

Effect-Size Measures

Beyond significance tests, quantify association magnitude. Key measures: **Cohen's d** (above) for mean differences; **Point-biserial correlation** $r_{pb} = d / \sqrt{d^2 + 4}$ relates d to a correlation-like effect size. **Odds Ratio (OR)** arises from logistic regression: $OR = e^\beta$ where β is the feature's log-odds coefficient. For a binary imaging biomarker and mutation outcome, $OR > 1$ implies higher odds of mutation in feature-positive

group. OR is scale-dependent (for continuous features, OR per unit or per SD). Confidence intervals on OR (via logistic regression) indicate precision. An OR of 2 implies the odds of mutation are doubled per unit increase; 95% CI not crossing 1 is “significant.” **Relative Risk (RR)** is similar but rarely estimated for case-control mutation data.

Assumptions: Cohen’s d and OR presume reasonably large, representative samples. Cohen’s d assumes normal-ish distributions; OR assumes linear log-odds relation. Both are **interpretable** on their scale.

Limitations: They ignore feature nonlinearity unless data are stratified; ORs from univariate logistic are confounded by omitted variables. Rare mutations may yield unstable ORs (wide CI, inflated OR). Radiomics often uses percentile transformations (e.g. z-score or binarization) to stabilize OR computations.

Classification Performance Metrics

When radiomic features are used in a predictive model of mutation status, **classification metrics** assess how well the model distinguishes mutation+ vs -. Common metrics (derived from a confusion matrix of true vs predicted labels) include: **Accuracy** = $(TP+TN)/N$; **Sensitivity (Recall)** = $TP/(TP+FN)$; **Specificity** = $TN/(TN+FP)$; **Precision (PPV)** = $TP/(TP+FP)$; **F1-score** = harmonic mean of precision and recall. However, with imbalanced classes (e.g. rare mutations) accuracy can be misleading.

ROC AUC (Area Under the Receiver Operating Characteristic curve) is widely used. AUC is the probability a randomly chosen mutation+ tumor is scored higher by the model than a mutation- one [27†L106-L114] . AUC=1.0 is perfect, 0.5 is random. It is threshold-independent and accounts for class imbalance well.

Interpretability: AUC is a rank-based measure, not an effect size per se, but a higher AUC means better discrimination. **Limitations:** AUC ignores calibration and actual probability values. It weights all thresholds equally and can be insensitive to clinically relevant regions of the ROC (e.g. high-specificity region).

Bootstrap confidence intervals: We recommend bootstrapping the test set for CIs. For example:

```
from sklearn.metrics import roc_auc_score
y_score = model.predict_proba(X_test)[:,-1]
auc = roc_auc_score(y_test, y_score)
# Bootstrap 95% CI
bootstraps = []
for i in range(1000):
    idx = np.random.choice(len(y_test), len(y_test), replace=True)
    if len(np.unique(y_test[idx]))<2: continue
    bootstraps.append(roc_auc_score(y_test[idx], y_score[idx]))
ci_lower = np.percentile(bootstraps, 2.5)
ci_upper = np.percentile(bootstraps, 97.5)
```

Precision-Recall (PR) curves and **PR AUC** are useful when positive class is rare (imbalanced). The **Matthews correlation coefficient (MCC)** is another robust metric for imbalanced classes (ranges -1 to 1, with 0 random) – see PLOS ONE comparisons. We cite standard references for AUC and ROC [27†L106-L114] , and note TRIPOD guidelines: always report discrimination (AUC) and calibration (e.g. calibration curve) for risk models.

Validation: Split data into training/testing or use cross-validation to estimate metrics reliably. Nested CV should be used if feature selection or hyperparameter tuning is performed to avoid optimistic bias. When comparing models, use DeLong's test or bootstrapped ROC curves to assess AUC differences.

Information-Theoretic Measures

Entropy and **mutual information (MI)** quantify dependence without assuming linearity. *Entropy* $H(X)$ of a feature X measures its randomness: $H = -\sum p(x) \log p(x)$. One can compute first-order entropy of a radiomic histogram (common in texture analysis). However, "entropy" can also mean *joint entropy* or *matrix-based entropy* (e.g. from a GLCM). Higher entropy indicates more heterogeneity. It has no fixed scale and is sensitive to binning choices.

Mutual Information (MI) between feature X and mutation label Y is

$$I(X; Y) = \sum_{x,y} p(x, y) \log \frac{p(x, y)}{p(x)p(y)}.$$

It captures any dependency (linear or nonlinear). $MI \geq 0$ (0 if independent). Estimation requires discretizing continuous features or using k-nearest neighbor estimators. Strength: model-free, can find associations missed by correlation. Limitation: high variance in finite samples and difficulty in interpreting magnitude. MI lacks an intuitive "effect-size" label (units of bits). **Example code (discrete case):**

```
from sklearn.feature_selection import mutual_info_classif
mi = mutual_info_classif(X, y, discrete_features=False)
```

or for two arrays `x, y`:

```
from sklearn.metrics import mutual_info_score
mi_xy = mutual_info_score(np.digitize(x, bins), y)
```

For **multiple features vs gene sets**, one can compute multivariate MI or conditional MI, but this scales poorly. Alternative: build joint decision trees or use **information gain** from entropy reduction in a decision tree.

Assumptions: None (nonparametric), but requires enough data to cover feature range. MI is also sensitive to identical confounding effects: if scanner id strongly influences both a feature and gene, MI will flag the association (confounded).

Multivariate and Model-Based Approaches

Logistic regression (for binary mutation outcome) or **multinomial/ordinal regression** (for >2 categories) are standard. Coefficients β_i for each feature give log-odds changes; exponentiating yields OR. Multivariate models allow adjusting for confounders (e.g. include tumor volume, age, scanner as covariates). Assumes

linear additive effects on log-odds, independent errors, no multicollinearity. In radiomics, often thousands of features exist: need dimension reduction or regularization. **Example (using statsmodels):**

```
import statsmodels.api as sm
X_aug = sm.add_constant(X_train) # adds intercept
logit = sm.Logit(y_train, X_aug).fit()
coef = logit.params
or_vals = np.exp(coef) # Odds ratios
ci = np.exp(logit.conf_int())
```

LASSO / Elastic Net regression: These penalized generalized linear models shrink/zero out weak features, useful for high-dimensional radiomics. The nonzero coefficients indicate important features. Interpretability: coefficients (log-odds) or ORs as above. Regularization requires tuning (via CV). **Strength:** handles $p \gg n$, does feature selection. **Limitation:** may randomly select among correlated features; effect sizes are biased downwards by penalty. Feature importance here is heuristic.

Tree-based models (Random Forest, XGBoost): These nonparametric classifiers capture nonlinear interactions. **Feature importance** can be derived (Gini importance or permutation importance). More interpretable: use **SHAP values** (SHapley Additive exPlanations) to quantify each feature's contribution to each prediction. SHAP provides local and global importance with additive feature attribution. **Formula:** SHAP values satisfy $f(x) = \phi_0 + \sum_i \phi_i(x)$. **Sensitivity:** Trees are less sensitive to monotonic confounders if included as features, but still can exploit spurious associations.

Example SHAP code (after fitting an RF/XGBoost):

```
import shap
explainer = shap.Explainer(model, X_train)
shap_values = explainer(X_test)
shap.summary_plot(shap_values, X_test)
```

(Requires `shap` library.)

Partial least squares (PLS) or Canonical Correlation Analysis (CCA): Link entire radiomic feature space to genomic profiles (e.g. gene expression). Rarely used for single-gene mutation, more for expression signatures. These yield multivariate associations but require large sample sizes.

Support Vector Machines (SVM): Often used with kernels for complex decision boundaries. Metrics as above (AUC) but interpretation is via support vectors (no natural effect size).

Assumptions/Notes: Multivariate models assume no strong collinearity (standardize features). They can accommodate confounder adjustment explicitly. **Robustness to high-dimensionality:** Regularized models (LASSO) and tree ensembles cope well; classical logistic without regularization fails if $p \gg n$. Effect sizes (e.g. OR) from multivariate models are *adjusted* for covariates (less confounded than univariate).

Outcome-Adjusted / Survival Analysis

Often radiogenomic studies also consider clinical outcomes. To assess associations **adjusted for survival or outcome**, one can use joint models or incorporate outcomes as covariates. For example, modeling mutation status *given* radiomics and survival jointly is complex. More common: **stratified analysis** (e.g. analyze only patients with similar stage).

Cox proportional hazards regression: Test if radiomic features or mutation status predict survival, adjusting for each other. Example metric: hazard ratio (HR) for each feature (e.g. increase in risk of death per feature unit) [42†L2775-L2784]. **Concordance index (C-index):** measures discrimination of survival model (analogous to AUC) [47†L79-L87] [45†L52-L60]. For a Cox model or Random Survival Forest, C-index = fraction of all usable patient pairs for which the model correctly orders survival times [47†L79-L87]. It accounts for censored data. Formula (Uno 2011) is given in [47†L83-L90]. C-index=1 is perfect, 0.5 random. **Limitations:** C-index ignores calibration and may be optimistic in small samples. Time-dependent AUC or integrated Brier score are also used for survival performance, but beyond scope.

For mutation association adjusted for prognosis, one could model mutation ~ radiomics + outcome (as reverse logistic), but this is uncommon. More typical: include mutation and radiomics together in a Cox model to see independent prognostic value.

Causal-Inference Methods

True causality (feature → gene mutation) is usually not inferable from purely observational radiogenomic data, since imaging is an effect not cause of genetics. However, one can attempt **causal modeling** to guard against confounding. For example:

- **Directed acyclic graphs (DAGs):** Prior to analysis, draw a causal graph of (Genomics ← Confounders → Radiomics) or (mutation → imaging). Identify needed covariates.
- **Instrumental variables or Mendelian randomization:** Use genetic variants as instruments to test if imaging features mediate their effect on outcome; not directly a “metric” but a methodology.
- **Conditional independence tests:** Test if association remains after conditioning on known confounders (e.g. tumor size, batch). If not, the association may be confounded.
- **Interventional data:** As [42] notes, causal claims require experimental validation (e.g. PDX models with gene editing) [42†L2775-L2784]. In silico, one may use matched cohorts or propensity score weighting (treat radiomics as “treatment” and mutation as “outcome” or vice versa) to reduce bias.

Metrics: No single “causal metric” like AUC exists; instead report adjusted effect measures (e.g. adjusted OR) and emphasize consistency across stratified analyses. Some advanced methods (e.g. Bayesian networks, structural equation models) could quantify directed associations, but these are rarely applied in radiomics due to complexity.

Sensitivity to Confounders (Scanner, Segmentation, Tumor Size)

All association metrics can be biased if *nonbiological factors* correlate with both features and gene status. In radiomics, scanner/protocol differences are notorious confounders. For example, intensity histograms differ by scanner model, affecting texture features [38†L193-L202]. **Harmonization:** Combat (empirical

Bayes) is widely used to remove batch effects from features before analysis [39†L85-L94] . It centers and scales features by batch (scanner) to a common distribution [39†L85-L94] . Even after harmonization, residual confounding may persist, so include scanner/site as covariates.

Segmentation variability: Different contouring can alter feature values. Robust features (e.g. shape, some intensity metrics) have high test-retest ICC; others (high-order textures) may vary greatly. We recommend calculating ICC on repeat segmentations and discarding unstable features before association tests [38†L193-L202] .

Tumor size and stage: Larger tumors often have more heterogeneity and may correlate with mutation burden or grade. This induces confounding: e.g. correlation between a texture metric and mutation may actually reflect size effects. Include tumor volume as covariate in multivariate tests or use size-normalized features (e.g. ratio metrics) to mitigate this.

Metric-specific sensitivity: As an example, Figure 2 below conceptually compares metric sensitivity to confounding (higher = more easily biased). Unadjusted univariate tests (Pearson r, t-test) are highly sensitive; regularized multivariate models are somewhat more robust if confounders are included; nonlinear metrics (MI, tree models) can also latch onto confounders if present. (This illustration is schematic.)

【1†embed_image】 *Figure: Hypothetical comparison of metrics' sensitivity to imaging confounders (scanner, segmentation). Bars (higher = more sensitive). Multivariate methods and covariate adjustment are needed to mitigate such biases.*

Preprocessing and Normalization

Image preprocessing: Intensity normalization (e.g. z-score, Nyul method for MRI) ensures features are comparable across patients. For CT, standardizing window/level or using Hounsfield units helps. For MRI, bias-field correction reduces intensity inhomogeneity. Co-registration or resampling to uniform voxel size (e.g. 1×1×1 mm) avoids scale issues in shape features.

Feature harmonization: After feature extraction, apply harmonization (ComBat or variants) to correct scanner effects [39†L85-L94] . Also remove features with low variance or near-zero ICC across test-retest scans.

Feature scaling: Most multivariate models require scaling (standardizing to mean 0, SD 1) so that coefficients and penalties are on the same scale. It also improves convergence of LASSO/SVM. For interpretable effect sizes, report OR per 1-SD increase.

Multiple testing correction: When hundreds of features are tested univariately, control FDR (Benjamini-Hochberg) or use Bonferroni. Alternatively, use feature selection first (filter by correlation) and then test the reduced set to lessen the multiple-comparison burden.

Validation Strategies

- **Cross-validation (CV):** Use k-fold CV or repeated CV on the *training* data for hyperparameter tuning (e.g. LASSO penalty, number of trees). Then have a held-out *test* set for final evaluation. Ensure that feature selection or scaling is performed *inside* the CV loop to avoid information leakage.
- **External validation:** Whenever possible, test the model/association in an independent cohort (different center). The radiogenomics review stresses that independent validation is crucial [【53†L5840-L5847】](#) .
- **Bootstrap/Permutation:** For single-metric confidence intervals and p-values, bootstrap resampling is recommended. For example, permute mutation labels to estimate null distributions of AUC or correlation.
- **Multiple-testing correction:** In discovery phases, control for the large feature space. In predictive modeling, adjust for multiple model comparisons.
- **Reporting standards:** Follow guidelines (e.g. TRIPOD, Radiomics Quality Score) by reporting effect sizes with CIs, discrimination (AUC/C-index) with CIs, and calibration (Hosmer–Lemeshow test, calibration curve if relevant).

Metric Comparison Table

Metric	Type	Effect Size?	Nonlinear?	High-dim Robust?	Use Case	Sample Size	Refs
<i>Pearson r</i>	Univariate	Yes (r)	No	No	Exploratory correlation	≥30 (per Corollary)	—
<i>Spearman ρ</i>	Univariate	Yes (ρ)	Monotonic	No	Nonlinear monotonic association	Moderate	—
<i>t-test / ANOVA</i>	Univariate	Yes (Cohen's d)	No	No	Compare feature means	≥20/ group	—
<i>Chi-square / Fisher</i>	Univariate	Yes (OR, φ)	No	No	Categorical features	≥50 total	—
<i>Mutual Information</i>	Univariate	No	Yes	No	Nonlinear dependence	Large (≫50)	—
<i>Logistic Regression</i>	Multivariate	Yes (OR, β)	Linear log-odds	No (nonreg.)	Predict mutation, adjust covariates	≥10 × features	—
<i>LASSO / Elastic Net</i>	Multivariate	Yes (sparse β)	Linear log-odds	Yes	High-dim feature selection	≥10 × features	—

Metric	Type	Effect Size?	Nonlinear?	High-dim Robust?	Use Case	Sample Size	Refs
<i>Random Forest</i>	Model-based	No (importance)	Yes	Moderately	Nonlinear classification, SHAP	Moderate	—
<i>SHAP Values</i>	Model/explain	No (coalitional)	Yes	No (derived)	Model interpretability	≥100 instances	Lundberg et al. (2017) ¹
<i>ROC AUC</i>	Performance	No	Indirectly	Moderately	Discrimination of classifier	≥50 per class	【27†L106-L114】
<i>Accuracy, F1, MCC</i>	Performance	No	Indirectly	No	Thresholded classification	≥50 per class	—
<i>Concordance Index</i>	Performance	No	Indirectly	No	Survival model discrimination	≥50	【47†L79-L87】 【45†L52-L60】
<i>Correlation Ratio η</i>	Univariate	Yes	Curvilinear	No	ANOVA effect (non-linear)	≥30/group	—
<i>Cox regression (HR)</i>	Multivariate	Yes (HR)	Linear hazard	No	Mutation vs survival analysis	≥10 events/var	—

Table: Metrics for radiomics–genomics association. “Type” indicates analysis context; “Effect size?” if the metric yields a direct effect-size. “Handles nonlinear?” if metric captures nonlinear association. Sample size rule-of-thumb: radiomics often requires large N because of high dimensionality. Referenced formulas: see text.

Example Metric Computations (Pseudocode)

- **Mutual Information:** For discretized feature `x` and binary mutation `y`:

```
from sklearn.metrics import mutual_info_score
# discretize continuous feature, e.g. 10 bins
bins = np.histogram_bin_edges(x, bins=10)
x_binned = np.digitize(x, bins)
mi = mutual_info_score(x_binned, y)
```

- **AUC with Bootstrap CI:** (as above) use `roc_auc_score` and bootstrap sampling.
- **Logistic Regression OR:** Fit logistic using statsmodels:

```
import statsmodels.api as sm
logit = sm.Logit(y, sm.add_constant(X)).fit()
```

```
coef = logit.params
or_vals = np.exp(coef)
ci = np.exp(logit.conf_int())
# OR and 95% CI for each feature
```

- **LASSO Feature Importance:** Using scikit-learn:

```
from sklearn.linear_model import LogisticRegressionCV
model = LogisticRegressionCV(penalty='l1', solver='saga', cv=5,
scoring='roc_auc')
model.fit(X_train, y_train)
coefs = model.coef_[0]
important = [f for f,c in zip(features, coefs) if c!=0]
```

- **SHAP values:** (after training `model` on data):

```
import shap
explainer = shap.Explainer(model, X_train)
shap_vals = explainer(X_test)
shap.summary_plot(shap_vals, X_test) # global importance
```

- **C-index for Survival:** Using a library (e.g. `lifelines` or `pysurvival`):

```
from lifelines.utils import concordance_index
c_index = concordance_index(event_times, -predicted_risk, event_observed)
```

(or formula in [【47†L83-L90】](#)).

Best-Practice Pipeline

1. **Data collection & QC:** Acquire imaging and matched genomic data. Ensure standardized imaging protocols (slice thickness, scanner) or record metadata for later adjustment [【53†L5844-L5847】](#) .
2. **Segmentation:** Delineate tumor (and possibly peritumoral) regions. Use consensus or automated methods with high reproducibility; compute ICC on a subset to retain robust features.
3. **Image preprocessing:** Normalize intensities (MRI bias correction, histogram matching; CT to standard HU windows) to reduce scanner effects.
4. **Feature extraction:** Compute radiomic features (shape, histogram, texture, wavelets) under consistent settings (e.g. fixed bin size) as per IBSI standards.
5. **Feature harmonization:** Adjust features for batch/scanner with ComBat or similar [【38†L193-L202】](#) . Remove near-zero-variance or highly collinear features. Optionally apply PCA or clustering to reduce dimensionality.
6. **Exploratory analysis:** Compute univariate metrics (correlation, t-tests, MI) between each feature and mutation. Correct for multiple tests (FDR). Examine effect sizes (Cohen's d, OR) for top hits.

7. **Multivariate modeling:** Build predictive models (e.g. LASSO-logistic, random forest) using selected features. Include clinical covariates (age, tumor size) to adjust. Use nested CV for feature selection and hyperparameter tuning.
8. **Validation:** Evaluate models on held-out data or external cohorts. Report AUC, accuracy, sensitivity/specificity, and C-index for survival if applicable, with CIs via bootstrap [53†L5844-L5847] . Perform permutation tests to assess significance under the null.
9. **Interpretation:** Analyze model coefficients or SHAP values to interpret feature–mutation links. Verify that associations are not driven by confounders (e.g. stratify by scanner or run a sensitivity analysis excluding potential confounders).
10. **Clinical integration:** If validated, integrate radiogenomic biomarkers into decision-support, ensuring calibration and decision thresholds are clinically meaningful.

Prioritized Metrics by Goal

- **Discovery (hypothesis generation):** Univariate *effect sizes* (Cohen’s d, OR) and nonparametric measures (MI, Spearman) to flag candidates. Use rigorous multiple-testing correction. Visualize associations (box plots, volcano plots of effect vs $-\log p$).
- **Predictive modeling:** Emphasize *classification metrics* (ROC AUC, F1, etc.) and model-derived importance (LASSO coefficients, SHAP). Use nested CV and external validation. Prefer balanced accuracy or MCC if classes are unbalanced. Report calibration (Brier score or calibration curves) alongside discrimination [53†L5844-L5847] .
- **Causal inference:** Use multivariate regression with confounder adjustment; consider DAGs to choose covariates. Sensitivity analysis (e.g. remove suspected confounders) and, if available, instrumental approaches. Ultimately, in silico findings should be validated by experiments (e.g. gene editing in PDX) [42†L2775-L2784] .

References: Key methods and practices are discussed in recent radiogenomics reviews [53†L5833-L5841] [38†L193-L202] and ML evaluation guides [27†L106-L114] [47†L79-L87] .
